

AgriRover: Design and Early Prototype of a Low-Cost Edge-AI Ground Rover for Horticultural Crop Scouting

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PRIMARY THEME

Theme V · Digital and Smart Agriculture

SUB-THEME

Precision Agriculture

ABSTRACT

Frequent, close-range crop scouting is difficult to sustain across smallholder horticulture farms, while aerial imaging can miss symptoms beneath dense canopies and may impose recurring service and operator costs. This paper presents the design and early prototype of AgriRover, a compact four-wheel platform intended for supervised-to-autonomous scouting of tomato fields in Maharashtra. The assembled prototype currently supports remote driving, operator-controlled emergency stopping, obstacle awareness, soil-sensing bench tests, and camera-based crop observation.

The proposed field architecture separates time-critical control from perception. An ESP32 controller manages drive, sensing, watchdogs, and fail-safe states; a Raspberry Pi 5 with optional edge acceleration performs local inference, evidence capture, and mission logging without requiring continuous cloud connectivity. A MobileNetV2/TFLite pipeline supports 38 PlantVillage crop-health classes. The navigation design combines range sensors, wheel encoders, an inertial measurement unit, and vision; GNSS is retained for coarse plot geotagging rather than crop-row selection. A two-way Telegram interface supports status, stop, and photo commands and is structured for image-linked, Marathi advisory messages.

Engineering verification includes CAD and electrical designs, a modular pilot bill of materials estimated at ₹41,150, simulation, and 146 passing automated software tests covering perception interfaces, navigation logic, telemetry, alerts, secure model updating, and fail-safe behavior. These results establish implementation readiness for supervised field evaluation, not agronomic efficacy. Planned trials will therefore measure row-tracking error, mission completion, energy use, per-class precision and recall on locally collected tomato images, and agreement with agronomist diagnoses. The work contributes a repairable, validation-oriented reference architecture for affordable ground-based crop scouting in Indian smallholder settings.

Keywords: precision agriculture; agricultural robotics; edge AI; crop scouting; disease classification; smallholder farming